

False detection rate control in time series coincidence detection

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Paper: *False positive control in time series coincidence detection*, [arXiv:2512.17372](https://arxiv.org/abs/2512.17372)

Intro: coincidence detection & time-shifting

LIGO detectors

Detecting gravitational waves from binary mergers
(black holes or neutron stars)

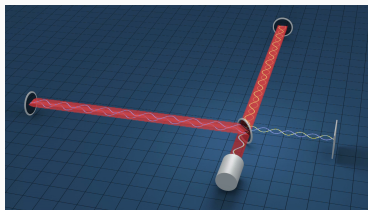
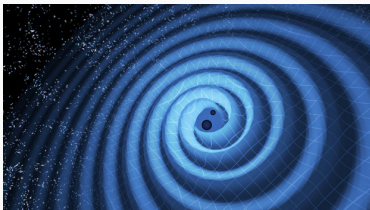


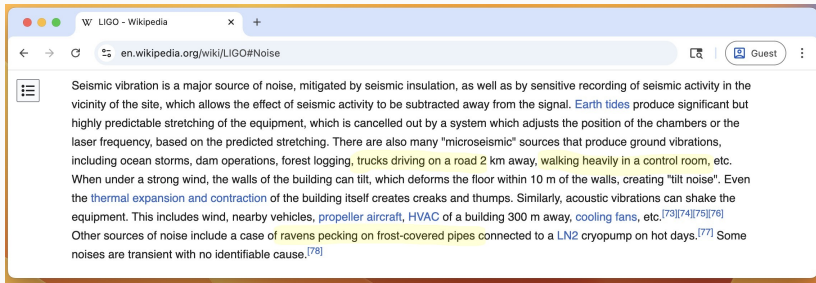
Image credit:

<http://www.ligo.caltech.edu/>

<https://en.wikipedia.org/wiki/LIGO>

Detecting gravitational waves with LIGO

How to separate signal from noise?



Source: <https://en.wikipedia.org/wiki/LIGO>

Detecting gravitational waves with LIGO

A key part of the solution:

Compare measurements across 2 (or more) detectors



LIGO Livingston



LIGO Hanford

Image credit: <http://www.ligo.caltech.edu/>

The coincidence detection problem

Setting: time series data

$$\mathbf{X} = (X_1, \dots, X_T), \quad \mathbf{Y} = (Y_1, \dots, Y_T)$$

Core question: do we observe shared trends in $\mathbf{X}$ and $\mathbf{Y}$
(e.g., shared spikes, overall synchronicity,)
beyond what might occur by random chance?

The coincidence detection problem

Example: Detecting gravitational waves with LIGO

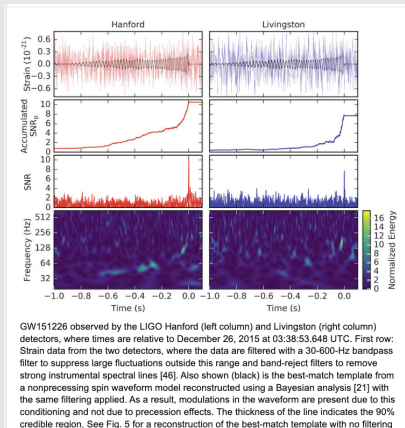


Image credit: *GW151226: Observation of Gravitational Waves from a 22-Solar-Mass Binary Black Hole Coalescence*, Abbott et al 2016

The coincidence detection problem

Example: Gravitational waves and gamma ray bursts

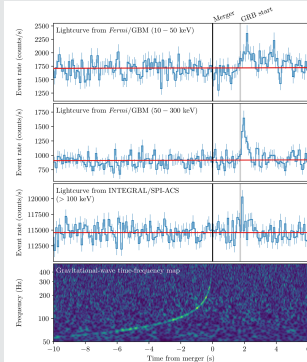


Figure 2. Joint, multi-messenger detection of GW170817 and GRB 170817A.

Top: the summed GBM lightcurve for sodium iodide (NaI) detectors 1, 2, and 5 for GRB 170817A between 10 and 50 keV, matching the 100 ms time bins of the SPI-ACS data. The background estimate from Goldstein et al. (2016) is overlaid in red. Second: the same as the top panel but in the 50–300 keV energy range. Third: the SPI-ACS lightcurve with the energy range starting approximately at 100 keV and with a high energy limit of at least 80 MeV.

Bottom: the time-frequency map of GW170817 was obtained by coherently combining LIGO-Hanford and LIGO Livingston data. All times here are referenced to the GW170817 trigger time T_0^{GW} .

Image credit: *Gravitational Waves and Gamma-rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A*, LIGO Scientific Collaboration et al 2017

The coincidence detection problem

Example: Synchronicity in neuron spike train data

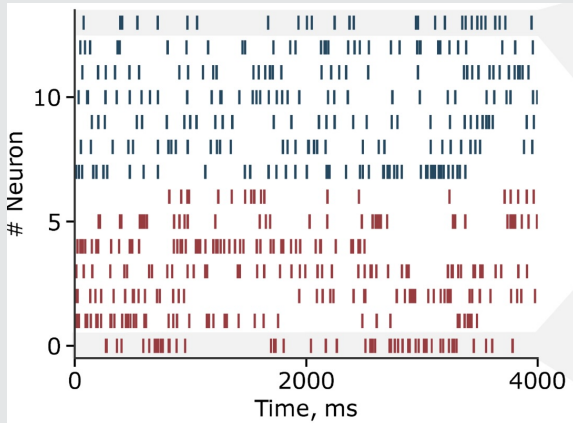


Image credit: *Spikebench: An open benchmark for spike train time-series classification*, Lazarevich et al 2023

The coincidence detection problem

Example: Comparing flood events and epidemic outbreaks

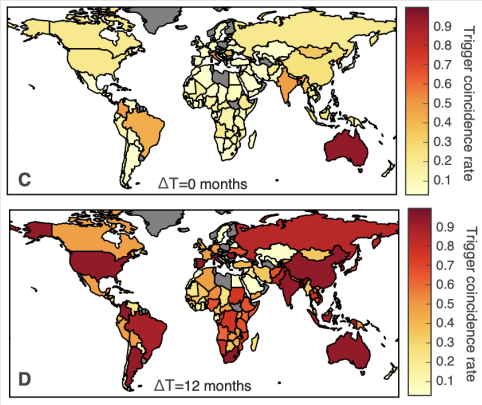


Image credit: *Event coincidence analysis for quantifying statistical interrelationships between event time series: On the role of flood events as triggers of epidemic outbreaks*, Donges et al 2016

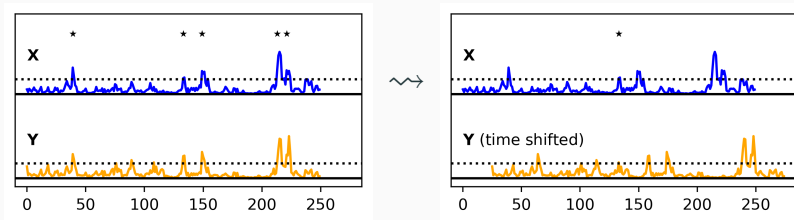
The time-shifting approach

Time series data

$$\mathbf{X} = (X_1, \dots, X_T), \quad \mathbf{Y} = (Y_1, \dots, Y_T)$$

Main idea behind time-shifting:

- Shift the data streams $\mathbf{X}$ and $\mathbf{Y}$ relative to each other
- Use the shifted data to estimate the false positive rate of a detection procedure



The time-shifting approach

Related to many statistical methods & approaches:

- Permutation tests of independence
(Pitman 1937; Fisher 1956; Lehmann et al 1986)
- Efron's work on permutation methods & bootstrap methods
(including applications to astro, e.g., Efron & Petrosian 1992)
- Permutation/bootstrap tests under time series dependence
(Politis & Romano 1991; Chernozhukov et al 2018; Romano & Tirlea 2022)
- Inference in time series under mixing conditions
(Chernozhukov et al 2021; Xu & Xie 2023; Oliveira et al 2024)

The time-shifting approach

Key question

Understand the statistical properties of the time-shifting method:

Does time-shifting offer finite-sample guarantees
to control the false positive rate, for time-series data?

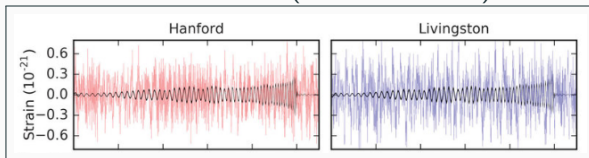
Detecting events vs. detecting synchronicity

Question type (1) : detecting events

Do the two data streams both indicate an event detection?

$$\begin{aligned}\mathbf{X} &= (X_1 \quad \dots \quad X_{t-1} \quad X_t \quad X_{t+1} \quad X_{t+2} \quad X_{t+3} \quad \dots \quad X_T) \\ \mathbf{Y} &= (Y_1 \quad \dots \quad Y_{t-1} \quad \underbrace{Y_t \quad Y_{t+1} \quad Y_{t+2}}_{\text{event detected at time } \approx t?} \quad Y_{t+3} \quad \dots \quad Y_T)\end{aligned}$$

- Gravitational waves (two LIGO sites)



- Earthquake (sensors at different locations)
- Physiological event (multiple wearable devices)
- Neuron response to external stimulus

Question type (1) : detecting events

Measure evidence of an event at time t via a test statistic

e.g., $\Psi = \mathbf{1}_{X_t \geq x_*, Y_t \geq y_*}$, or $\Psi = \sqrt{X_t^2 + Y_t^2}$

More generally:

$$\Psi = \psi((X_t, \dots, X_{t+\Delta}), (Y_t, \dots, Y_{t+\Delta}))$$

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More generally:

$$\Psi = \psi((X_t, \dots, X_{t+\Delta}), (Y_t, \dots, Y_{t+\Delta}))$$

The time-shifting approach: compare Ψ against

$$\Psi_{i,j} = \psi((X_i, \dots, X_{i+\Delta}), (Y_j, \dots, Y_{j+\Delta})) \quad \leftarrow i, j \in [T - \Delta]$$

Question type (2) : detecting synchronicity

Do the two data streams show synchronicity overall?

$$\begin{array}{rcl} \mathbf{X} & = & (X_1 \quad X_2 \quad X_3 \quad \dots \quad X_{T-2} \quad X_{T-1} \quad X_T) \\ \mathbf{Y} & = & (Y_1 \quad Y_2 \quad Y_3 \quad \dots \quad Y_{T-2} \quad Y_{T-1} \quad Y_T) \end{array}$$

synchronous behavior across entire time range?

- Co-occurring events (predator/prey , flood/epidemic ,)
- Neurons that are connected to each other



Question type (2) : detecting synchronicity

Measure evidence of synchronicity:

$$\Psi = \psi((X_1, \dots, X_T), (Y_1, \dots, Y_T))$$

e.g., $\Psi = \sum_{t=1}^T \mathbf{1}_{X_t \geq x_*, Y_t \geq y_*}$

Question type (2) : detecting synchronicity

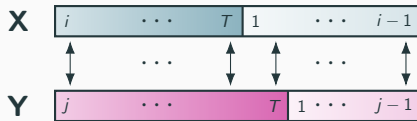
Measure evidence of synchronicity:

$$\Psi = \psi((X_1, \dots, X_T), (Y_1, \dots, Y_T))$$

$$\text{e.g., } \Psi = \sum_{t=1}^T \mathbf{1}_{X_t \geq x_*, Y_t \geq y_*}$$

The time-shifting approach: compare Ψ against

$$\Psi_{i,j} = \psi((X_i, \dots, X_T, X_1, \dots, X_{i-1}), (Y_j, \dots, Y_T, Y_1, \dots, Y_{j-1})) \quad \leftarrow i, j \in [T]$$



Detecting events vs. detecting synchronicity

Detecting events:

- **X** & **Y** are independent measurements of the same signal (e.g., LIGO)
- Validate detections by comparing **X** & **Y** against each other

Detecting synchronicity:

- **X** & **Y** measure two different signals (e.g., neurons)
- We aim to determine if these two signals are correlated

Detecting events vs. detecting synchronicity

Both problems:

- Large values of $\Psi = \psi(\dots)$ indicates evidence of a detection
- Time-shifting $\rightsquigarrow$ compute “control” values $\Psi_{i,j}$

Key question:

does time-shifting control the rate of false positive detections?

no real events / no synchronicity

Detecting events vs. detecting synchronicity

Both problems:

- Large values of $\Psi = \psi(\dots)$ indicates evidence of a detection
- Time-shifting $\rightsquigarrow$ compute “control” values $\Psi_{i,j}$

Key question:

does time-shifting control the rate of false positive detections?

under [^]what assumptions

no real events / no synchronicity

The time-shifting approach for detecting events

The time-shifted p-value

$$\begin{array}{lcl} \mathbf{X} & = & (X_1 \quad \dots \quad X_{t-1} \quad X_t \quad X_{t+1} \quad X_{t+2} \quad X_{t+3} \quad \dots \quad X_T) \\ \mathbf{Y} & = & (Y_1 \quad \dots \quad Y_{t-1} \quad \underbrace{Y_t \quad Y_{t+1} \quad Y_{t+2}}_{\text{event detected at time } \approx t?} \quad Y_{t+3} \quad \dots \quad Y_T) \end{array}$$

Test statistic:

$$\Psi = \psi((X_t, \dots, X_{t+\Delta}), (Y_t, \dots, Y_{t+\Delta}))$$

Time-shifted test statistic:

$$\Psi_{i,j} = \psi((X_i, \dots, X_{i+\Delta}), (Y_j, \dots, Y_{j+\Delta}))$$

The time-shifted p-value

$$\begin{aligned}\mathbf{X} &= (X_1 \quad \dots \quad X_{t-1} \quad X_t \quad X_{t+1} \quad X_{t+2} \quad X_{t+3} \quad \dots \quad X_T) \\ \mathbf{Y} &= (Y_1 \quad \dots \quad Y_{t-1} \quad \underbrace{Y_t \quad Y_{t+1} \quad Y_{t+2}}_{\text{event detected at time } \approx t} \quad Y_{t+3} \quad \dots \quad Y_T)\end{aligned}$$

Test statistic:

$$\Psi = \psi((X_t, \dots, X_{t+\Delta}), (Y_t, \dots, Y_{t+\Delta}))$$

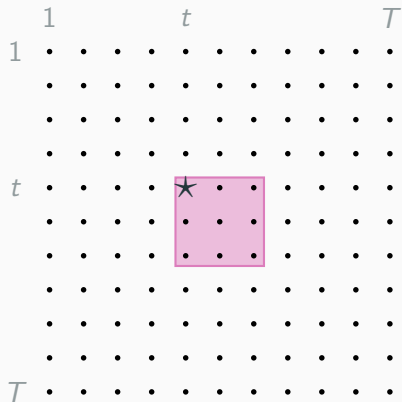
Time-shifted test statistic:

$$\Psi_{i,j} = \psi((X_i, \dots, X_{i+\Delta}), (Y_j, \dots, Y_{j+\Delta}))$$

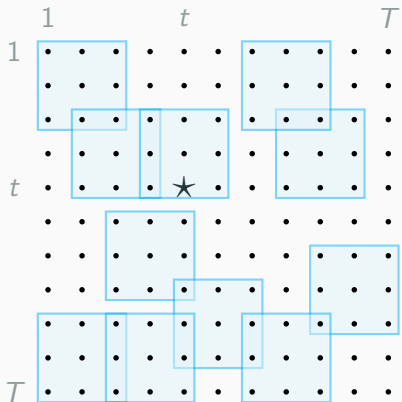
Time-shifted p-value

$$p_t = \frac{1}{(T - \Delta)^2} \sum_{i,j=1}^{T-\Delta} \mathbb{1}\{\Psi_{i,j} \geq \Psi\}$$

The time-shifted p-value



Ψ



$\Psi_{i,j}$

$i, j \in [T - \Delta]$

The time-shifted p-value

Time-shifted p-value

$$p_t = \frac{1}{(T - \Delta)^2} \sum_{i,j=1}^{T-\Delta} \mathbb{1}\{\Psi_{i,j} \geq \Psi\}$$

To prove that $\mathbb{P}\{p_t \leq \alpha\} \lesssim \alpha$ under the null,
typically would need to assume:

(1) $\Psi_{i,j} \stackrel{d}{=} \Psi$ under the null

and

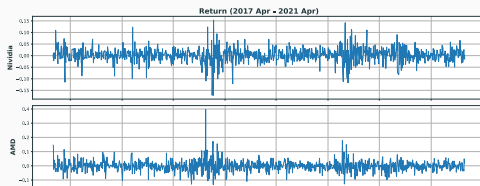
(2) Empirical distrib. of $\Psi_{i,j}$'s converges to true distribution

The time-shifted p-value

Assumption (2):

Empirical distrib. of $\Psi_{i,j}$'s converges to true distribution

Problem: the $\Psi_{i,j}$'s may have strong dependence



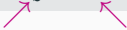
Stock returns data from: Gibbs & Candès 2021,

<https://github.com/igibbs/DtACI/tree/main/MajorDailyReturns>

The time-shifted p-value

Assumption (1): $\psi_{i,j} \stackrel{d}{=} \psi$ under the null

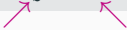
$\psi((X_i, \dots, X_{i+\Delta}), (Y_j, \dots, Y_{j+\Delta}))$ $\psi((X_t, \dots, X_{t+\Delta}), (Y_t, \dots, Y_{t+\Delta}))$



The time-shifted p-value

Assumption (1): $\psi_{i,j} \stackrel{d}{=} \psi$ under the null

$\psi((X_i, \dots, X_{i+\Delta}), (Y_j, \dots, Y_{j+\Delta}))$ $\psi((X_t, \dots, X_{t+\Delta}), (Y_t, \dots, Y_{t+\Delta}))$



Sufficient conditions for this to hold: under the null,

- $\mathbf{X}$ is a stationary time series
- $\mathbf{Y}$ is a stationary time series
- $\mathbf{X} \perp\!\!\!\perp \mathbf{Y}$

What can we prove about p_t under only these conditions?

Theoretical guarantee under stationarity

Theorem 1

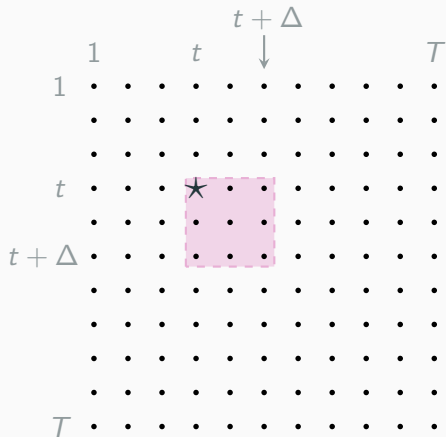
Assume $\mathbf{X}$ and $\mathbf{Y}$ are each stationary time series.

Then, if $\mathbf{X} \perp\!\!\!\perp \mathbf{Y}$,

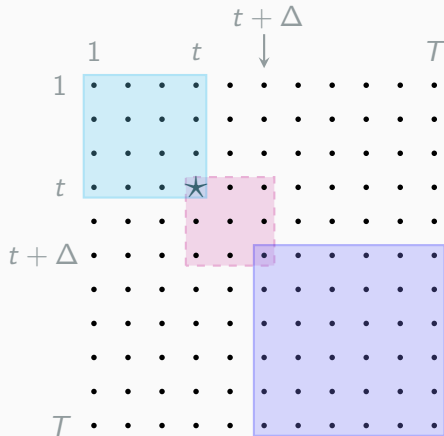
$$\mathbb{P}\{p_t \leq \alpha\} \leq \frac{1}{\text{mar}(t)} \cdot \alpha$$


$$= \left(\frac{\min\{t, T+1-t-\Delta\}}{T-\Delta} \right)^2$$

Theoretical guarantee under stationarity

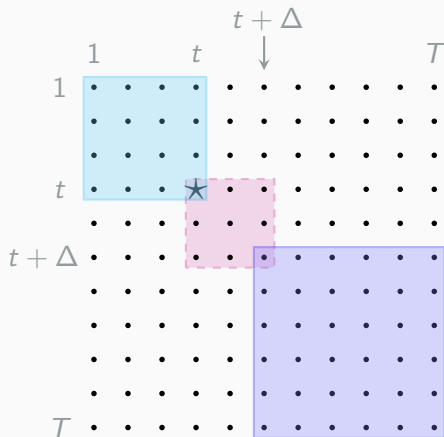


Theoretical guarantee under stationarity



$$\text{mar}(t) = \frac{\min \left\{ \text{light blue square}, \text{purple square} \right\}}{(T - \Delta)^2}$$

Theoretical guarantee under stationarity

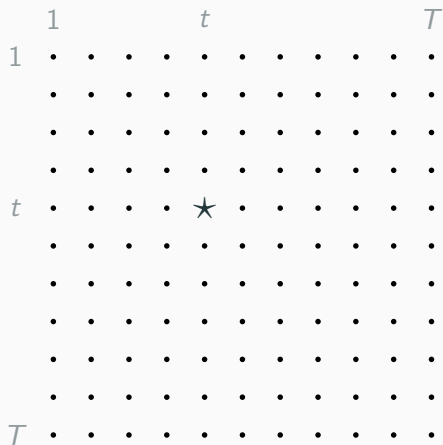


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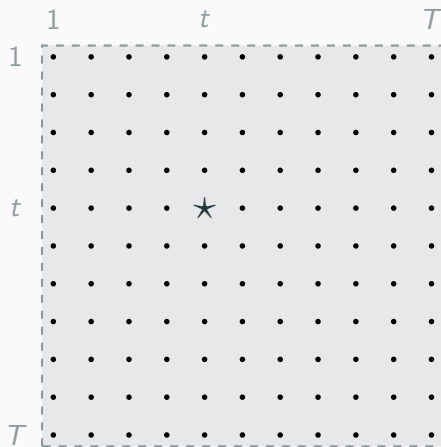
Example:

$\Delta = 0, t = T/2 \implies$ Theorem 1 implies $\mathbb{P}\{p_t \leq \alpha\} \leq 4\alpha$

Proof sketch — Theorem 1 (for the case $\Delta = 0$)

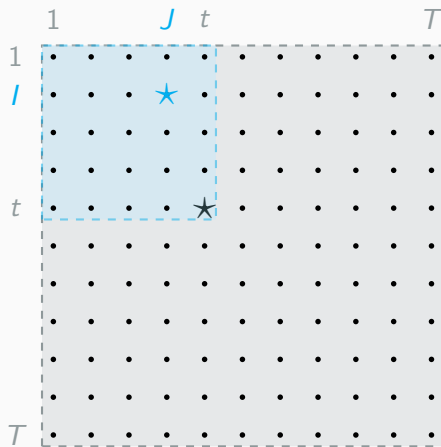


Proof sketch — Theorem 1 (for the case $\Delta = 0$)



$$p_t = \frac{1}{T^2} \sum_{i,j=1}^T \mathbb{1}\{\Psi_{i,j} \geq \Psi\}$$

Proof sketch — Theorem 1 (for the case $\Delta = 0$)



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$$\tilde{p}_{I,J} = \frac{1}{t^2} \sum_{i,j=1}^t \mathbb{1}\{\Psi_{i,j} \geq \Psi_{I,J}\}$$

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Define $\tilde{p}_{I,J} = \frac{1}{t^2} \sum_{i,j=1}^t \mathbb{1}\{\Psi_{i,j} \geq \Psi_{I,J}\}$ where $I, J \stackrel{\text{iid}}{\sim} \text{Unif}([t])$

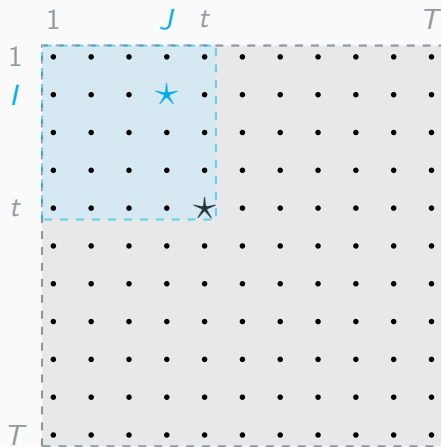
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Then $\mathbb{P}\{\tilde{p}_{I,J} \leq a\} \leq a$ for all $a \in [0, 1]$ because:

- $\Psi_{I,J}$ is a random draw from $(\Psi_{i,j})_{i,j \in [t]}$
- $\tilde{p}_{I,J}$ computes a p-value using empirical distrib. of $(\Psi_{i,j})_{i,j \in [t]}$

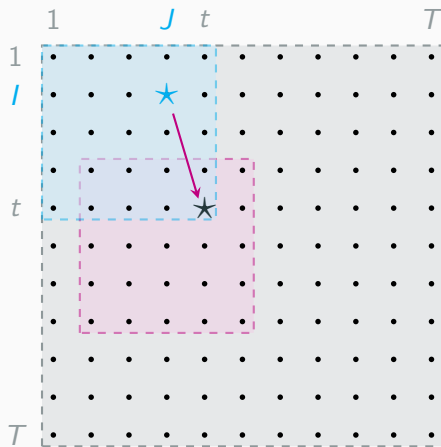
Proof sketch — Theorem 1 (for the case $\Delta = 0$)



$$p_t = \frac{1}{T^2} \sum_{i,j=1}^T \mathbb{1}\{\Psi_{i,j} \geq \Psi\}$$

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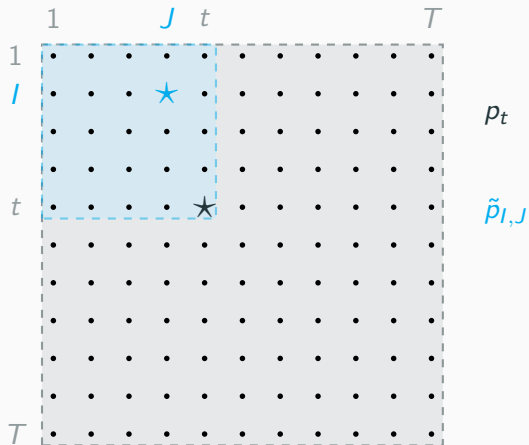
Proof sketch — Theorem 1 (for the case $\Delta = 0$)

$$\text{Define } \tilde{p}_{I,J} = \frac{1}{t^2} \sum_{i,j=1}^t \mathbb{1}\{\Psi_{i,j} \geq \Psi_{I,J}\} \text{ where } I, J \in [T]$$

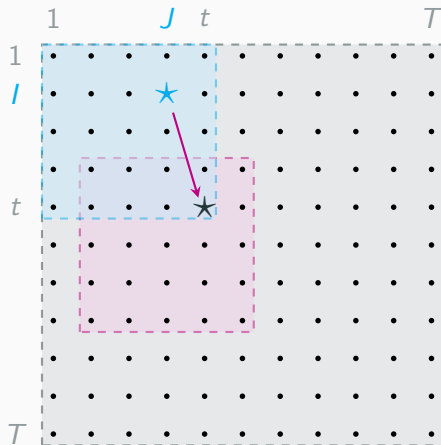
$$\text{Define } \tilde{p}_{I,J}^* = \frac{1}{t^2} \sum_{i=t+1-I}^{2t-I} \sum_{j=t+1-J}^{2t-J} \mathbb{1}\{\Psi_{i,j} \geq \Psi\} \text{ where } I, J \in [T]$$

$\tilde{p}_{I,J} \stackrel{d}{=} \tilde{p}_{I,J}^*$ under the null, since $\mathbf{X}, \mathbf{Y}$ are stationary & independent

Proof sketch — Theorem 1 (for the case $\Delta = 0$)



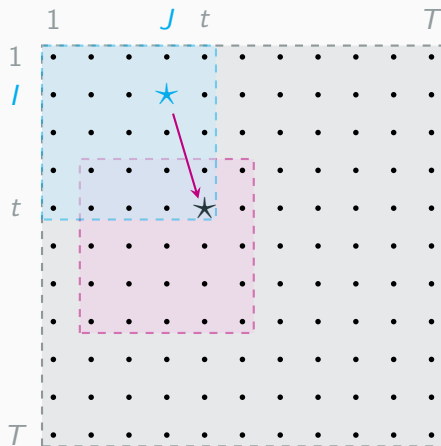
Proof sketch — Theorem 1 (for the case $\Delta = 0$)



$$p_t \geq \text{mar}(t) \cdot \tilde{p}_{I,J}^*$$

$$\tilde{p}_{I,J} \stackrel{\text{d}}{=} \tilde{p}_{I,J}^*$$

Proof sketch — Theorem 1 (for the case $\Delta = 0$)



$$p_t \geq \text{mar}(t) \cdot \tilde{p}_{I,J}^*$$

$$\tilde{p}_{I,J} \stackrel{d}{=} \tilde{p}_{I,J}^*$$

$$\mathbb{P} \{p_t \leq \alpha\} \leq \mathbb{P} \{\text{mar}(t) \cdot \tilde{p}_{I,J}^* \leq \alpha\} = \mathbb{P} \{\text{mar}(t) \cdot \tilde{p}_{I,J} \leq \alpha\} \leq \frac{\alpha}{\text{mar}(t)}$$

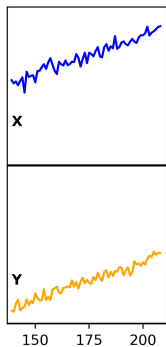
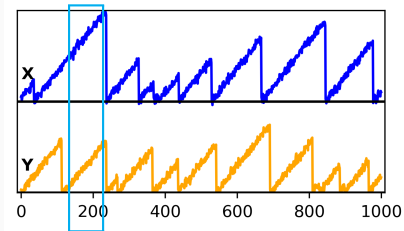
Theoretical guarantee under stationarity: worst case

Theorem 1 proves that:

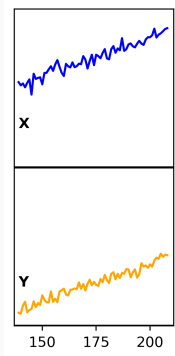
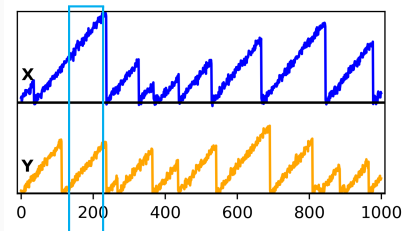
- If testing for an event at time $t \approx T/2$,
false detection inflation is at most $\mathcal{O}(1)$
- But if testing at time $t \approx T$, can be arbitrarily high

$\leadsto$ no meaningful result for real-time detections

Theoretical guarantee under stationarity: worst case



Theoretical guarantee under stationarity: worst case



Worst-case example for p_T

Under the above assumptions,
may have

$$\mathbb{P} \left\{ p_T = \frac{1}{T^2} \right\} \approx 1$$

Theoretical guarantee under stationarity: worst case

Theorem 1.5 (matching lower bound)

Assume $\Delta = 0$ and $\sqrt{\alpha}T$ is an integer.

There exist stationary time series $\mathbf{X} \perp\!\!\!\perp \mathbf{Y}$, and a test statistic ψ , such that

$$\mathbb{P}\{p_t \leq \alpha\} \geq \frac{\alpha}{\text{mar}(t)} \cdot \left(\frac{1}{1 + \sqrt{\frac{\alpha}{\text{mar}(t)}}} \right)^2$$

Theoretical guarantee under stationarity + β -mixing

Need to assume *mixing* to enable real-time inference:

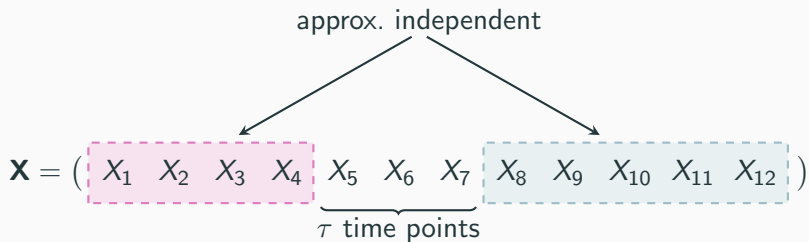
β -mixing coefficients

Let $\mathbf{X}'$ be an independent copy of $\mathbf{X}$. Then, for any k ,

$$d_{TV}((X_1, \dots, X_k, X_{k+\tau+1}, \dots, X_T), (X_1, \dots, X_k, X'_{k+\tau+1}, \dots, X'_T))$$

is at most $\beta_{\mathbf{X}}(\tau)$.

Theoretical guarantee under stationarity + β -mixing



Theoretical guarantee under stationarity + β -mixing

Theorem 2

Assume $\mathbf{X}$ and $\mathbf{Y}$ are each stationary time series.

Then, if $\mathbf{X} \perp\!\!\!\perp \mathbf{Y}$,

$$\mathbb{P}\{p_t \leq \alpha\} \leq \alpha + \frac{4(\tau + \Delta)}{T - \Delta} + 2\beta_{\mathbf{X}}(\tau) + 2\beta_{\mathbf{Y}}(\tau)$$

Theoretical guarantee under stationarity + β -mixing

Theorem 2

Assume $\mathbf{X}$ and $\mathbf{Y}$ are each stationary time series.

Then, if $\mathbf{X} \perp\!\!\!\perp \mathbf{Y}$,

$$\mathbb{P}\{p_t \leq \alpha\} \leq \alpha + \frac{4(\tau + \Delta)}{T - \Delta} + 2\beta_{\mathbf{X}}(\tau) + 2\beta_{\mathbf{Y}}(\tau)$$

Related problem: conformal predictive in time series

*Predictive inference for time series: why is
split conformal effective despite temporal dependence?*
(with Ashwin Pananjady)



Proof sketch — Theorem 2 (for the case $\Delta = 0$, $t = T$)

The p-value for real-time testing ($t = T$):

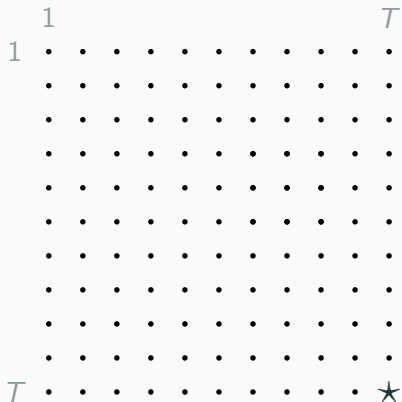
$$p_T = \frac{1}{T^2} \sum_{i,j=1}^T \mathbb{1}\{\Psi_{i,j} \geq \Psi\}$$

Compare to:

$$p_{I,J} = \frac{1}{T^2} \sum_{i,j=1}^T \mathbb{1}\{\Psi_{i,j} \geq \Psi_{I,J}\}$$

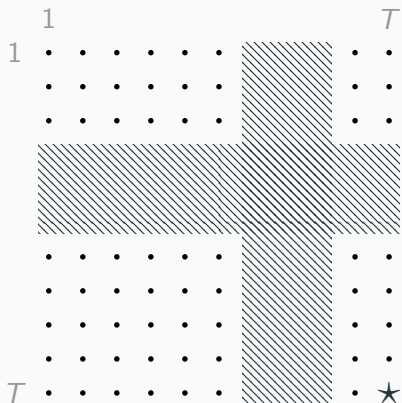

$$\mathbb{P}\{p_{I,J} \leq \alpha\} \leq \alpha \text{ for } I, J \stackrel{\text{iid}}{\sim} \text{Unif}([T])$$

Proof sketch — Theorem 2 (for the case $\Delta = 0$, $t = T$)



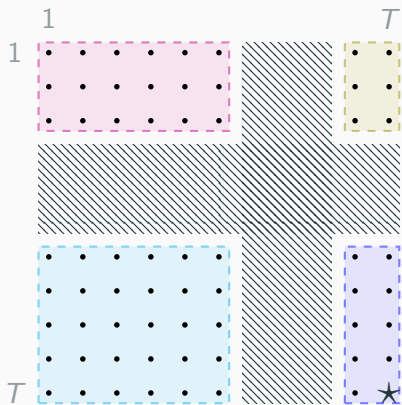
p_T

Proof sketch — Theorem 2 (for the case $\Delta = 0$, $t = T$)



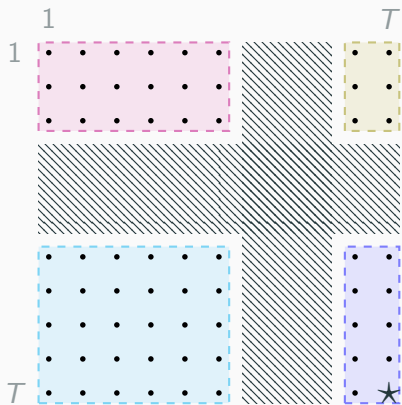
$$\approx p_T$$

Proof sketch — Theorem 2 (for the case $\Delta = 0$, $t = T$)

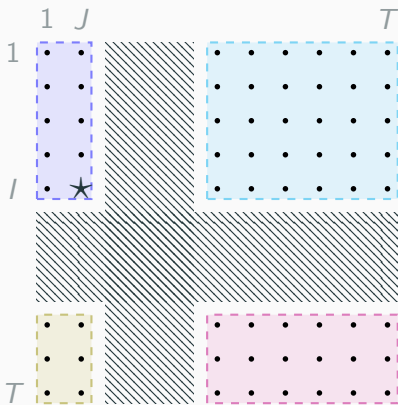


$$\approx p_T$$

Proof sketch — Theorem 2 (for the case $\Delta = 0$, $t = T$)



$\approx p_T$



$\approx p_{I,J}$

The multiple testing problem

Results so far:

- Declare an event at time t if $p_t \leq \alpha$
- $\rightsquigarrow$ Prob. of a false positive is $\lesssim \alpha$

If testing for events at each time $t \in \mathcal{T}$, use Bonferroni correction:

Declare an event at time t if $p_t \leq \alpha/|\mathcal{T}|$

The multiple testing problem

Extending Theorem 1

$$\mathbb{P} \left\{ \min_{t \in \mathcal{T}} p_t \leq \alpha / |\mathcal{T}| \right\} \leq \frac{1}{\underbrace{\text{mar}(\mathcal{T})}_{\text{harmonic mean of } \{\text{mar}(t)\}_{t \in \mathcal{T}}}} \cdot \alpha$$

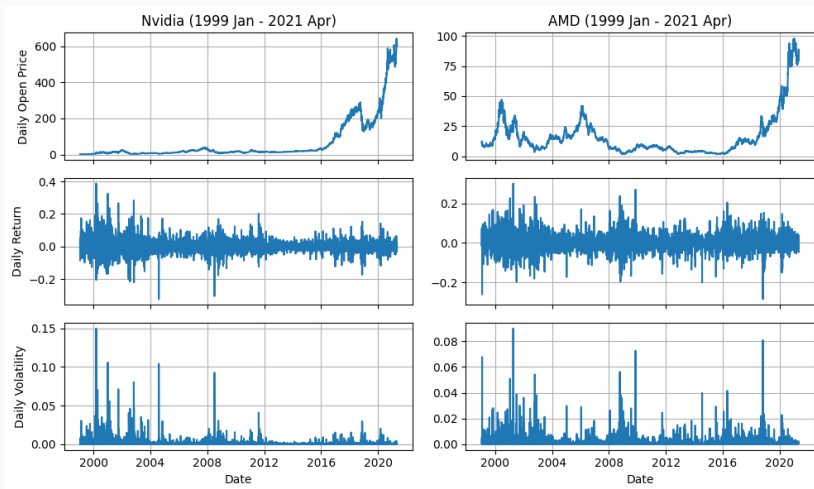
harmonic mean of $\{\text{mar}(t)\}_{t \in \mathcal{T}}$

Extending Theorem 2

$$\mathbb{P} \left\{ \min_{t \in \mathcal{T}} p_t \leq \alpha / |\mathcal{T}| \right\} \leq \alpha + \mathcal{O} \left(\sqrt{\alpha} \cdot \sqrt{\frac{\tau + \Delta}{T - \Delta}} \right) + 2\beta_X(\tau) + 2\beta_Y(\tau)$$

Empirical results

Real data: stock price of Nvidia and AMD



Stock returns data from: Gibbs & Candès 2021,
<https://github.com/isgibbs/DtACI/tree/main/MajorDailyReturns>

Experiment setting

- **Data streams $(\mathbf{X}, \mathbf{Y})$:**

- price pairs $(P_t^{\text{Nvidia}}, P_t^{\text{AMD}})$
- return pairs $(R_t^{\text{Nvidia}}, R_t^{\text{AMD}})$ $[R_t = \frac{P_t - P_{t-1}}{P_{t-1}}]$
- volatility pairs $(V_t^{\text{Nvidia}}, V_t^{\text{AMD}})$ $[V_t = R_t^2]$

Experiment setting

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- volatility pairs $(V_t^{\text{Nvidia}}, V_t^{\text{AMD}})$ $[V_t = R_t^2]$

- **Real data:**

Sample $T = 100$ consecutive time points from (Nvidia,AMD)

- **Synthetic null:**

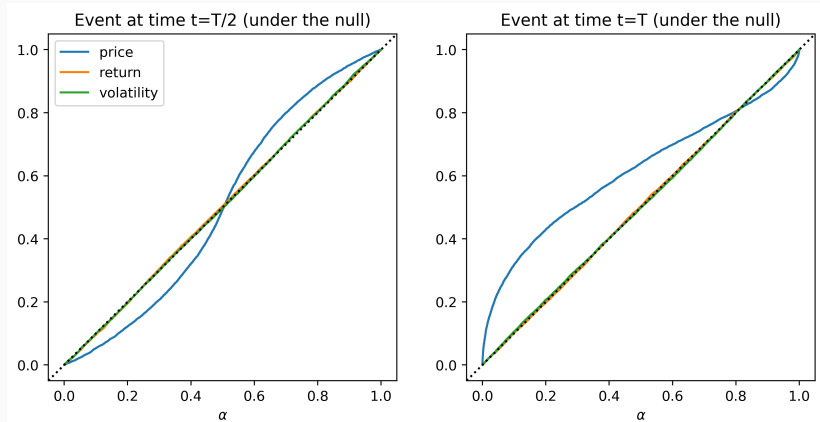
Sample $T = 100$ consecutive time points from Nvidia,
and $T = 100$ consecutive time points from AMD

- **Test statistic:**

$$\psi(X_t, Y_t) = X_t + Y_t$$

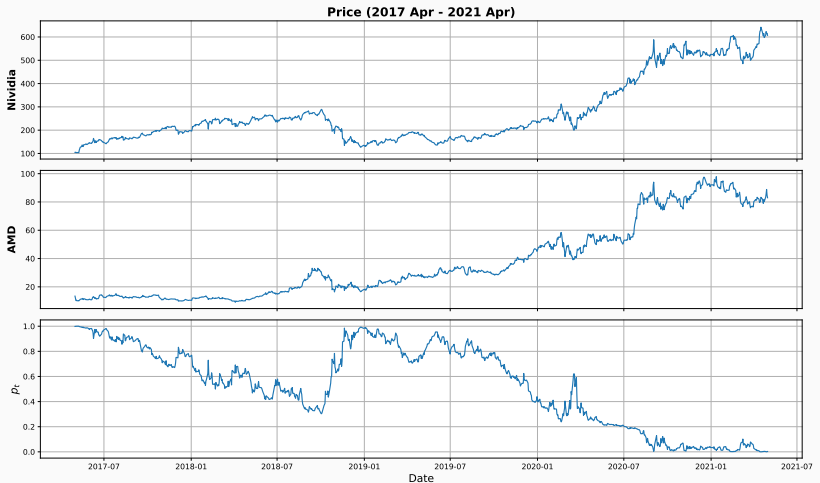
Results: event detection

Under the synthetic null:



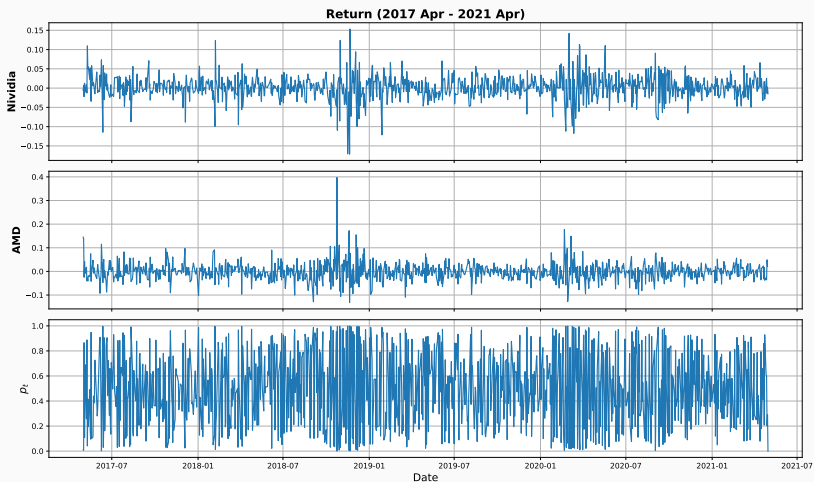
Results: event detection

For the real data:



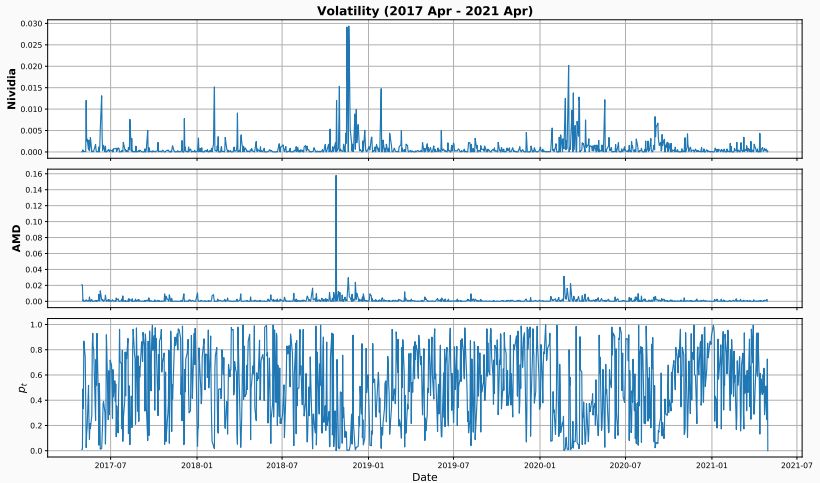
Results: event detection

For the real data:



Results: event detection

For the real data:



The time-shifting approach for detecting synchronicity

Detecting synchronicity

$$\begin{aligned}\mathbf{X} &= (X_1 \quad X_2 \quad X_3 \quad \dots \quad X_{T-2} \quad X_{T-1} \quad X_T) \\ \mathbf{Y} &= (Y_1 \quad Y_2 \quad Y_3 \quad \dots \quad Y_{T-2} \quad Y_{T-1} \quad Y_T)\end{aligned}$$

synchronous behavior across entire time range?

Test statistic:

$$\Psi = \psi((X_1, \dots, X_T), (Y_1, \dots, Y_T))$$

Compare to:

$$\Psi_{i,j} = \psi((X_i, \dots, X_T, X_1, \dots, X_{i-1}), (Y_j, \dots, Y_T, Y_1, \dots, Y_{j-1}))$$

The p-value:

$$p = \frac{1}{T^2} \sum_{i,j=1}^T \mathbf{1}\{\psi_{i,j} \geq \Psi\}$$

Intuition:

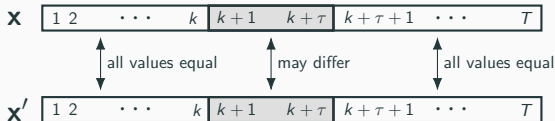
- Want to compare Ψ against its null distribution—
does $\psi_{i,j} \stackrel{d}{=} \Psi$ hold (approximately) under the null?

$$\psi_{i,j} = \psi((X_i, \dots, X_{T-1}, \textcolor{violet}{X}_T, \textcolor{violet}{X}_1, X_2, \dots, X_{i-1}), (Y_j, \dots, Y_{T-1}, \textcolor{violet}{Y}_T, \textcolor{violet}{Y}_1, Y_2, \dots, Y_{j-1}))$$

The stability condition

For any two sequences $\mathbf{x}, \mathbf{x}'$ of length T ,

$\mathbf{x} \stackrel{\tau}{\asymp} \mathbf{x}'$ means they only differ in τ many consecutive entries:



Stability coefficients for ψ

If $\mathbf{x} \stackrel{\tau}{\asymp} \mathbf{x}'$ and $\mathbf{y} \stackrel{\tau}{\asymp} \mathbf{y}'$ then $|\psi(\mathbf{x}, \mathbf{y}) - \psi(\mathbf{x}', \mathbf{y}')| \leq \gamma(\tau)$

Theorem 3

Assume ψ satisfies the stability condition with coefficients $\gamma(\tau)$.

Define a more conservative p-value

$$p^{+\varepsilon} = \frac{1}{T^2} \sum_{i,j=1}^T \mathbf{1}\{\psi_{i,j} + \varepsilon \geq \psi\},$$

where $\varepsilon \geq 4\gamma(\tau)$.

Then, for stationary $\mathbf{X} \perp\!\!\!\perp \mathbf{Y}$,

$$\mathbb{P}\{p^{+\varepsilon} \leq \alpha\} \leq \alpha + 2\beta_{\mathbf{X}}(\tau) + 2\beta_{\mathbf{Y}}(\tau).$$

Experiment: detecting synchronicity

- **Data streams ($\mathbf{X}, \mathbf{Y}$):**

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- **Real data:**

Sample $T = 100$ consecutive time points from (Nvidia,AMD)

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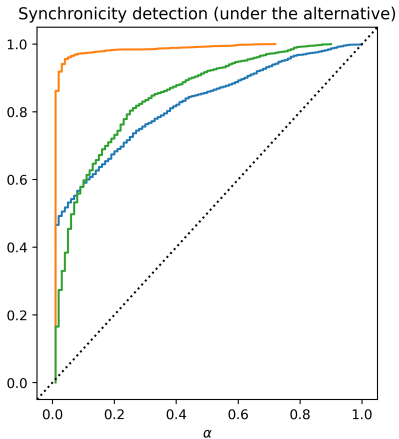
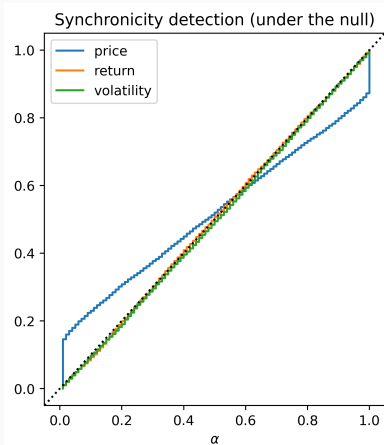
Sample $T = 100$ consecutive time points from Nvidia,
and $T = 100$ consecutive time points from AMD

- **Test statistic:**

$$\psi(\mathbf{X}, \mathbf{Y}) = \text{corr}(\mathbf{X}, \mathbf{Y})$$

Experiment: detecting synchronicity

Under the synthetic null:



Discussion

Summary & open questions

Our results offer non-asymptotic bounds on false positive rate for time-shifting methods, allowing for dependent data

To apply results to signal detection in astro—next steps:

- How to handle detections in a streaming setting?
- How to allow for the function $\psi(\dots)$ to evolve over time?
- Violations of stationarity—drift, seasonal effects, etc?

Thank you!